

Physiology of the digestive system



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WPS Office تعديل من خلال



Saliva production and secretion



General compositions of fluids

- Water
- Organic substances (proteins.....
- Inorganic substances (sodium, potassium, chlorine....)
- pH monitor (protons or bicarbonate)



Saliva composition

- Saliva (fluid secreted by the salivary glands for the purpose of chemical digestion)

Composition:

Water 99%

- Organic substances 0.3%
- Inorganic substances (Na^+ , K^+ , Cl^- ,....) 0,5%
- Bicarbonate



Organic substances:

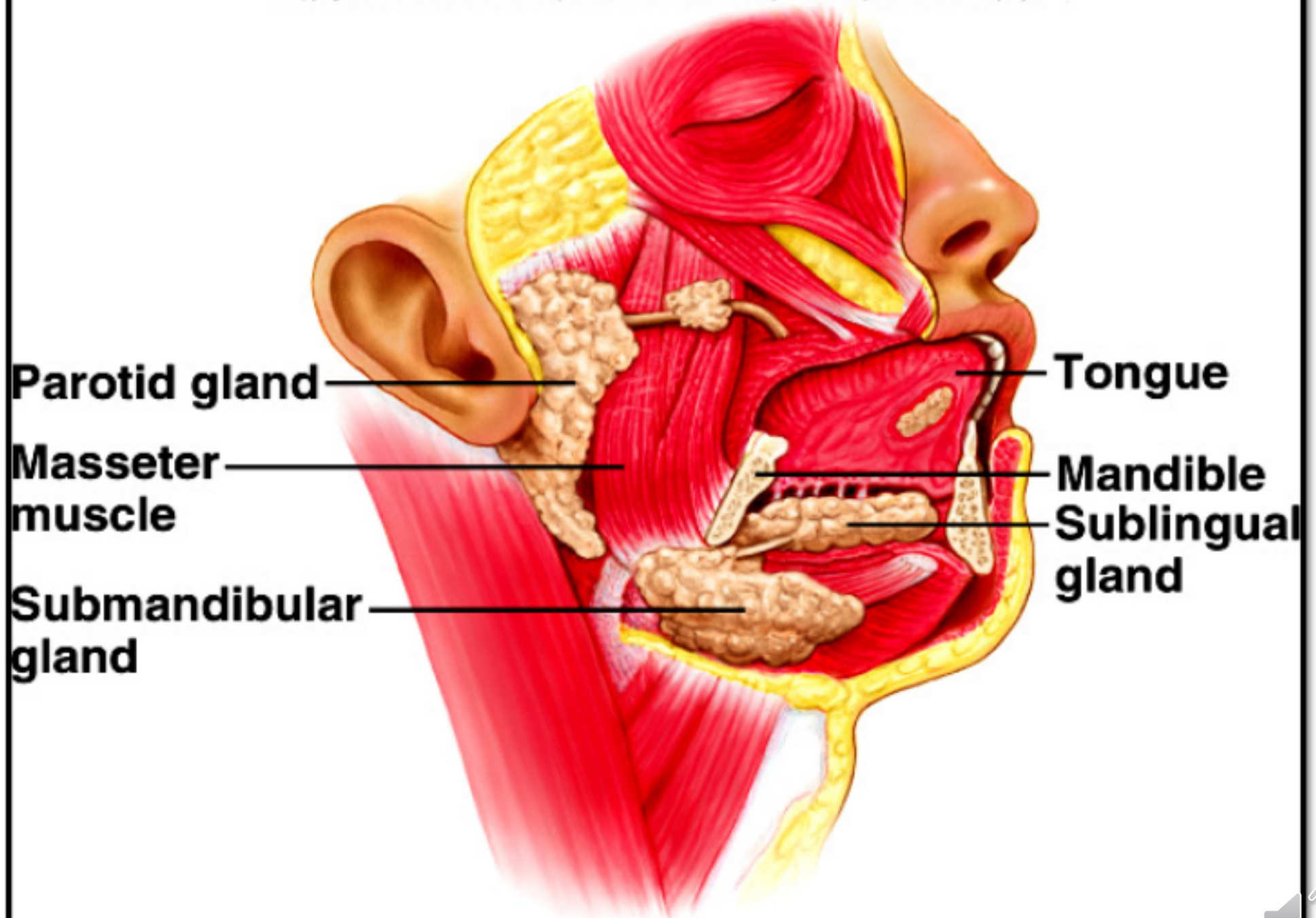
- Mucin,
- Digestive enzymes – ptyalin, lingual lipase,
- Proteolytic enzymes, cytochromoxidase, car anhydrase, phosphatase, IgA, lysozyme,
- Blood groups s...



The salivary glands are

- The parotid glands
- The submandibular gland
- The sublingual glands
- Buccal





Functions of the saliva

- 1-**Protection:** by lysosome & antibodies.
- 2- **Digestion:** by alpha amylase & lingual lipase.
- 3-**Lubrication (for the easement) of:** swallowing, speech & taste.
- 4-**Buffer:** neutralizes acids & relives heartburn.



The process of salivation

- Basal secretion is about 800 ml and increases to 1500 ml with the intake of food

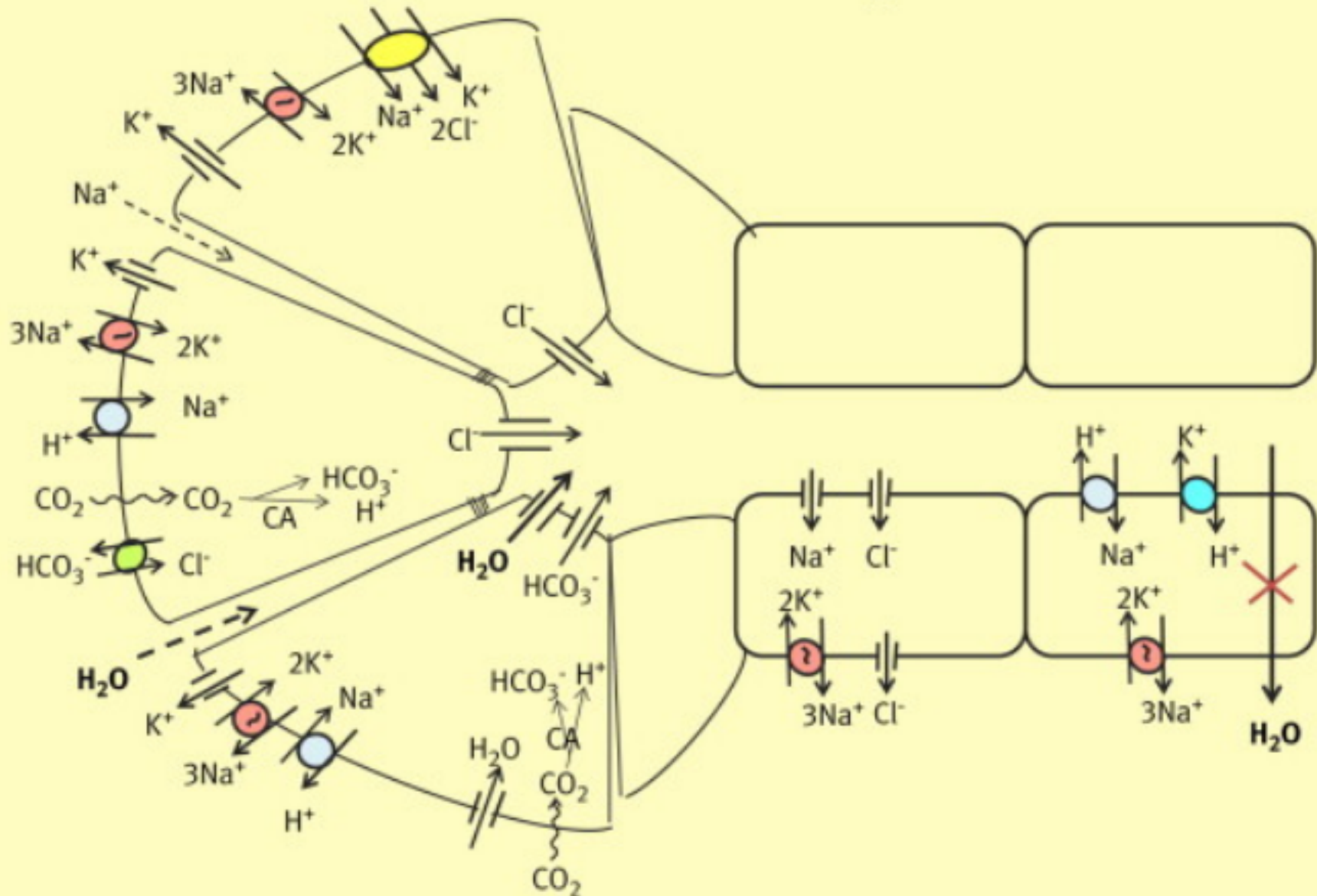


Acini

Primary secretion
- isotonic fluid -

Duct

Secondary ductal modification
- hypotonic fluid -

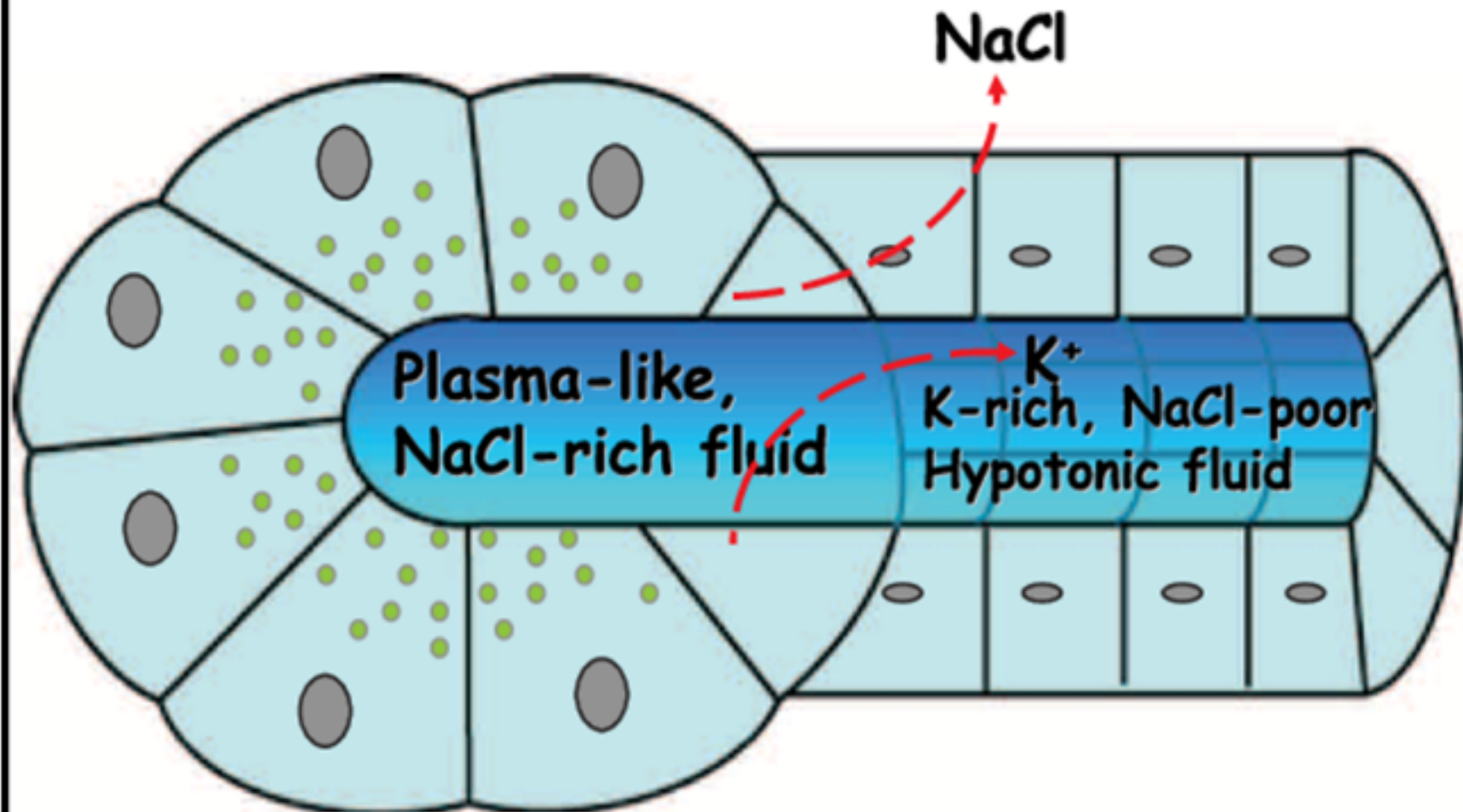


Fluid Secretion

**NaCl Reabsorption &
K⁺ Secretion**

Stage 1

Stage 2



Regulation

□ Nervous:

- Sympathetic increases mucous secretion
- Parasympathetic increases serous secretion
- **Centres:** salivatory nuclei (at the juncture of the medulla and pons)



Deficient salivation = xerostomia

